

Secure AI Insights Assistant

Overview

Build a secure AI-powered internal analytics assistant that can answer business questions using multiple private data sources.

The system should combine:

- Structured data from a relational database
- Unstructured data from PDF reports / internal documents
- CSV / spreadsheet business data

The assistant must provide useful answers while maintaining strong privacy and access boundaries.

This assignment is designed to evaluate:

- AI engineering capability
- Backend engineering
- Frontend engineering
- Tool/function calling architecture
- Multi-source retrieval systems
- Data handling maturity
- Security thinking
- Code quality
- Deployment readiness

Business Scenario

A fictional entertainment company stores internal business data across multiple systems.

Leadership wants a smart assistant that can answer questions like:

- Which movies performed best this year?
- Why is a specific title trending?
- Which genre is growing fastest?

- What audience segments are most engaged?
- What actions should leadership take next quarter?

The assistant must use approved internal tools and trusted retrieval methods.

Core Requirement

The system must work with **multiple data sources** and combine them intelligently.

Required Sources

Source A — SQL Database

Use provided relational data.

Source B — PDF Documents

Use provided internal reports.

Source C — CSV / Excel Files

Use provided business data files.

Mandatory Rules

1. Security

Sensitive internal data must be protected.

Your architecture should demonstrate safe handling of private information.

2. Tool-Based Access

The AI assistant should use backend tools / functions / services rather than unrestricted raw data access.

3. Explainability

Responses should indicate what sources were used when appropriate.

Functional Requirements

Backend

Build APIs / services for:

- data ingestion
- querying structured data
- document retrieval
- AI orchestration
- analytics generation

Use any backend stack such as:

- *Python + FastAPI*

Frontend

Build a usable interface with:

- Chat assistant UI
- Filters / selectors
- Insights panel
- Charts / visual summaries
- Query history or tool trace

Use any framework such as:

- *React*
- Next.js
- Vue

AI Layer

Use any model provider:

- OpenAI
- Anthropic
- *local model*
- Ollama

The model should help answer business questions using your architecture.

Required Example Questions

Your system should be able to answer questions like:

1. Which titles performed best in 2025?
 2. Why is Stellar Run trending recently?
 3. Compare Dark Orbit vs Last Kingdom.
 4. Which city had the strongest engagement last month?
 5. What explains weak comedy performance?
 6. What recommendations would you give for leadership?
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Provided Datasets

1. SQL / CSV Structured Data

Create random csv data files:

- movies.csv
- viewers.csv
- watch_activity.csv
- reviews.csv
- marketing_spend.csv
- regional_performance.csv

Load these into your chosen database or use directly.

2. PDF Documents

Create random demo/use any pdfs for the same:

- Quarterly executive report
- Campaign performance summary

- Content roadmap
- Policy guidelines
- Audience behavior report

Use them as additional knowledge sources.

Minimum Features

Required

- Working backend and frontend
- Multi-source answers
- At least one chart or visual summary
- Clean README
- Setup instructions

Strongly Recommended

- Clean project structure
 - Logging
 - Error handling
 - Validation
 - Reusable code
 - Containerized setup using Docker
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Submission Requirements

Please submit:

1. GitHub repository link
2. Setup instructions
3. Detailed README with architecture overview diagram
4. Notes on assumptions / tradeoffs

5. (Optional) 3 - 5 minute demo video

Evaluation Criteria

Category	Weight
Architecture Quality	25
Backend Engineering	20
AI / Multi-source Reasoning	20
Frontend Experience	15
Security / Data Handling	10
Code Quality	5
Documentation	5

What We Appreciate

- *Working prototype*
 - Thoughtful architecture
 - Clean APIs
 - Practical UX
 - Strong reasoning
 - Reliable outputs
 - Clear communication
 - Good engineering judgment
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Notes

You are free to make reasonable assumptions where details are not explicitly specified.

Please document those assumptions clearly.

We are interested not only in the final output, but also in how you think and structure systems.

Final Reminder

Build something practical, secure, and maintainable.

Perfect polish is less important than strong engineering choices.